

PROJECT REPORT

Import Data using Transform Maps (Spreadsheet)

Bulk employee data migration into ServiceNow using Import Sets, Transform Maps and Coalesce, with Reports and a Dashboard

Project Name	Import Data using Transform Maps (Spreadsheet)
Platform	ServiceNow (Global application scope)
Team ID	SWTID-2026-2605
Team Members	Monikasri M K – Team Leader
Version	1.0

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1. INTRODUCTION

1.1 Project Overview

Organisations regularly receive bulk employee data as Excel or Google Sheets files from HR teams and external systems. Typing these records into ServiceNow one by one is slow and error-prone, and importing the same or an updated file a second time can create duplicate records. This project solves that problem with ServiceNow's built-in data import framework.

The spreadsheet data (Employee ID, Name, Email, Department and Location) is first loaded into an **Import Set table** that acts as a staging area. A **Transform Map** then maps the staged source fields to the fields of a custom target table, **Employee Test (u_employee_test)**. **Coalesce** is enabled on Employee ID so that a repeated import updates matching records, inserts only new employees and ignores unchanged rows. The result of every load is verified in Transform History, and the final data is presented through three reports combined on the **Employee Analytics Dashboard**.

The project simulates a real-world onboarding and update scenario and helps beginners understand the end-to-end import process: data loading, field mapping, transformation execution, duplicate prevention and result verification.

Item	Detail
Source data	Sample Spreadsheet.xlsx (Google Sheets export), 15 employee rows, 5 columns
Staging (Import Set) table	Employee Import (u_employee_import)
Target table	Employee Test (u_employee_test) with 5 String fields
Transform Map	Sample Spreadsheet Import (Employee Import to Employee Test)
Duplicate control	Coalesce = true on Employee ID
Reports	Employees by Department (Pie), Employees by Location (Bar), Employee List Report (List)
Dashboard	Employee Analytics Dashboard

Table 1: Project at a glance

1.2 Purpose

The purpose of the project is to provide a reliable, repeatable and duplicate-safe way of moving spreadsheet data into ServiceNow, and to give HR and administrators a clear view of the imported data. The specific objectives are to:

- Load structured employee data from an Excel spreadsheet into a ServiceNow staging table using Load Data.
- Create a custom target table (Employee Test) and map source fields to it with a Transform Map.
- Prevent duplicate records by enabling Coalesce on the Employee ID field, so repeated imports update rather than duplicate.
- Validate every import using the Progress page, Transform History and the Employee Test list.

- Build reports (pie, bar, list) and combine them on a dashboard that can be shared with users, groups and roles.

2. IDEATION PHASE

2.1 Problem Statement

A customer problem statement helps the team view the problem from the user's point of view. The statement below was written for the main user, the ServiceNow administrator.

I am	A ServiceNow administrator who receives bulk employee data (Employee ID, Name, Email, Department, Location) in an Excel spreadsheet.
I'm trying to	Import that data into the ServiceNow platform quickly and accurately, and keep it up to date when new spreadsheets arrive.
But	Typing records one by one is slow, and importing the same or an updated file again creates duplicate or incorrect records.
Because	There is no staging area, field mapping or unique-key check (Coalesce) to validate data before it reaches the target table.
Which makes me feel	Frustrated, stressed about data accuracy, and unsure whether the records in ServiceNow can be trusted.

Table 2: Customer problem statement (PS-1)

PS	I am	I'm trying to	But	Because	Which makes me feel
PS-1	ServiceNow Administrator	Import bulk employee data from an Excel spreadsheet accurately	Manual entry is slow and re-importing creates duplicates	No staging, field mapping or Coalesce check	Frustrated, worried about data accuracy
PS-2	HR Manager / Reporting Analyst	See employee counts by department and location	Data is scattered across spreadsheets with no reliable view	Records are not centralised with reports or a dashboard	Uninformed, unable to decide with confidence

Table 3: Problem statements

2.2 Empathy Map Canvas

The empathy map captures what the administrator thinks, hears, sees, says and does, together with the pains and gains that shape the solution.

User / Persona: ServiceNow Administrator (HR data admin) who receives bulk employee data in Excel for onboarding and updates and must load it accurately without duplicates.	
Think & Feel • "Will this data land correctly the first time?" • Worried about duplicate or wrong employee records • Under pressure to onboard new joiners on time • Wants a repeatable process that is not error-prone	Hear • Manager: "New joiners' records are needed by Monday." • HR: "The spreadsheet is updated again, please refresh ServiceNow." • Colleagues: "Manual entry always leads to typos." • Community: "Use Import Sets and Transform Maps for bulk loads."

See • Excel sheets with dozens or hundreds of employee rows • The same Employee ID in different files with changed names or emails • Records typed one by one into ServiceNow forms • Import logs showing inserts, updates and errors	Say & Do • Says: “Can we automate this instead of typing each record?” • Cross-checks rows against Excel • Re-uploads corrected files and checks Transform History • Builds reports to show HR the final numbers
Pain • Slow, repetitive manual data entry • Duplicate records when the same file is imported twice • Typos and inconsistent names, emails and departments • Hard to prove what was inserted and what was updated • No quick view of employees by department or location	Gain • Whole spreadsheet loaded in one go through an Import Set • Field mapping done once in a reusable Transform Map • Coalesce on Employee ID updates instead of duplicating • Transform History confirms inserts, updates and ignored rows • Dashboard gives HR live visibility of employee data

Table 4: Empathy map

Key insight: the administrator needs a repeatable, duplicate-safe way to bring spreadsheet data into ServiceNow and to see the results clearly. This leads to the solution: Import Set (staging) + Transform Map (field mapping) + Coalesce on Employee ID + Reports and a Dashboard.

2.3 Brainstorming

The team gathered to discuss how bulk employee data received as an Excel spreadsheet can be moved into ServiceNow quickly and accurately. Ground rules were: volume over value, no idea is a bad idea, build on each other’s ideas, and stay focused on problem statement PS-1. Twelve ideas were listed and grouped into four themes: Data Loading, Mapping & Transformation, Data Integrity and Reporting.

Each idea was scored on **Importance** (value to the project goal) and **Feasibility** (how easily it can be built in ServiceNow) on a 1 to 5 scale. Ideas in the high-importance, high-feasibility area were selected.

#	Idea	Group	Imp.	Feas.	Decision
1	Load Excel data into an Import Set using Load Data	A: Data Loading	5	5	Selected
2	Accept CSV / TSV files	A: Data Loading	3	5	Future scope
3	Scheduled imports from an external Data Source	A: Data Loading	4	2	Future scope
4	Custom target table (u_employee_test)	B: Mapping	5	4	Selected
5	Transform Map with Auto Map Matching Fields	B: Mapping	4	5	Selected
6	Transform scripts to clean data	B: Mapping	3	4	Future scope
7	Coalesce on Employee ID (no duplicates)	C: Integrity	5	4	Selected
8	Verify loads with Transform History & Import Log	C: Integrity	4	4	Selected
9	Mandatory-field validation rules	C: Integrity	3	2	Future scope

#	Idea	Group	Imp.	Feas.	Decision
10	Reports: by Department, by Location, Employee List	D: Reporting	4	4	Selected
11	Employee Analytics Dashboard	D: Reporting	4	3	Selected
12	Scheduled / shared reports	D: Reporting	2	3	Future scope

Table 5: Idea scoring and decision

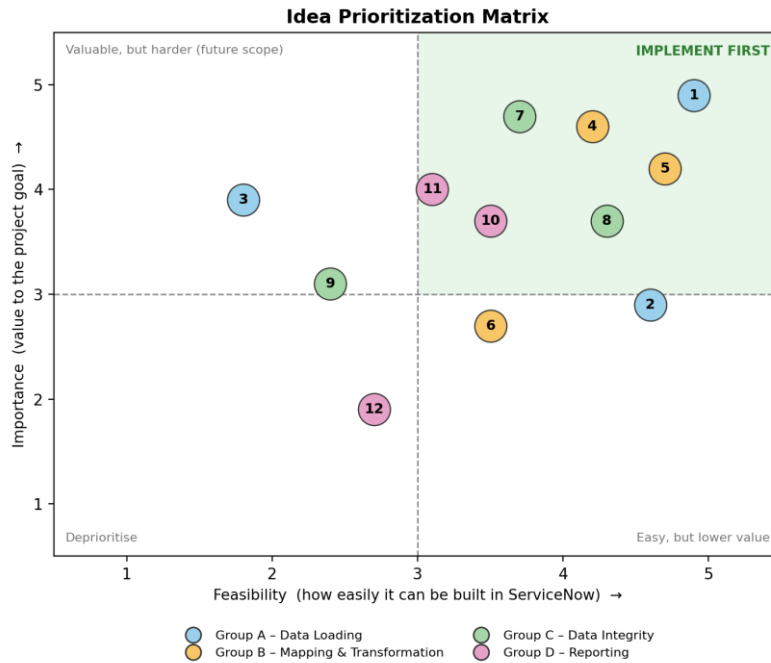


Figure 1: Importance versus feasibility matrix of the 12 ideas

Outcome: ideas 1, 4, 5, 7, 8, 10 and 11 form the final solution. The remaining ideas are recorded as future enhancements (Section 10).

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

The journey map follows the administrator from receiving the spreadsheet to presenting the results to HR, and shows where the solution improves the experience.

Stage	User activity	Touchpoint	Experience / pain point	Opportunity addressed
1. Receive data	Receives the employee list as an Excel or Google Sheet	Email, Google Sheets	Worried about volume and deadline	Prepare the sheet once and export as .xlsx
2. Prepare target	Creates the Employee Test table and its five String fields	Tables, Form Layout	Unsure how to structure the data	Simple custom table with matching fields
3. Load data	Uploads the file with Load Data into the staging table	System Import Sets, Load Data	Fear of changing live data directly	Staging table keeps raw data separate
4. Map and transform	Creates the Transform Map, auto-maps fields and runs Transform	Transform Map, Mapping Assist	Manual entry is slow and typo-prone	Bulk transform in about a second
5. Validate and de-duplicate	Enables Coalesce, re-imports changed data, reviews Transform History	Transform History, Import Log	Duplicates and unclear results	Coalesce updates or ignores; counts prove it
6. Report	Builds three reports and a dashboard, shares with HR	Reports, Dashboards, Share	Needs a quick view of headcount	One shared dashboard with live charts

Table 6: Customer journey map

3.2 Solution Requirement

Functional requirements

FR No.	Requirement (Epic)	Sub requirements
FR-1	Data Preparation	Prepare employee data in a spreadsheet and download it as .xlsx Create the custom table Employee Test (u_employee_test) Create Employee ID, Employee Name, Email, Department, Location
FR-2	Data Import (Import Set)	Upload the spreadsheet through System Import Sets > Load Data Create the staging table Employee Import (u_employee_import) Load later files into the existing Import Set table
FR-3	Field Mapping & Transformation	Create the Transform Map "Sample Spreadsheet Import" Map fields using Auto Map Matching Fields / Mapping Assist Run the Transform and confirm State Complete and Success
FR-4	Duplicate Prevention (Coalesce)	Enable Coalesce on the Employee ID field map Update existing records when the Employee ID matches Insert only new IDs and ignore unchanged records

FR No.	Requirement (Epic)	Sub requirements
FR-5	Validation & Verification	View imported records in the Employee Tests list Review Transform History (total, inserts, updates, ignored, skipped, errors) Check the Import Log for errors
FR-6	Reports	Employees by Department (Pie, Count) Employees by Location (Bar, Count) Employee List Report (List, 5 columns)
FR-7	Dashboard & Sharing	Create the Employee Analytics Dashboard Add the three reports Share with users, groups and roles

Table 7: Functional requirements

Non-functional requirements

NFR No.	Requirement	Description
NFR-1	Usability	Guided, menu-based screens (Load Data, Transform Map, Mapping Assist) with no coding. The target table has five simple String fields and list columns can be personalised.
NFR-2	Security	Access is controlled by roles and ACLs. Administrator access is needed for tables, Transform Maps and the dashboard form (pa_dashboards create ACL with Admin overrides). The dashboard can be restricted to selected users, groups and roles.
NFR-3	Reliability	Raw data stays in a separate staging table, every load reports a State and Completion code, and Coalesce keeps the employee table consistent across repeated imports.
NFR-4	Performance	Bulk processing: 15 rows in about 1.4 seconds and a 4-row update file in about 0.04 seconds.
NFR-5	Availability	Runs on the cloud-hosted ServiceNow instance through a web browser, so no local installation is needed.
NFR-6	Scalability	Import Sets and Transform Maps are built for bulk loads. The same map can be reused for new spreadsheets and extra fields, tables or reports can be added.

Table 8: Non-functional requirements

3.3 Data Flow Diagram

The data flow diagram shows how data enters the system, what transforms it and where it is stored.

Ref.	Name	Description
1.0	Load Data (Import Set)	The Excel file is uploaded through System Import Sets > Load Data and its rows are stored in the staging table.
2.0	Transform Map (field mapping)	The Transform Map maps source fields to target fields using Auto Map Matching Fields / Mapping Assist.
3.0	Coalesce Check (Employee ID)	Each incoming ID is compared with existing records: a new ID is inserted, a matching ID is updated and an unchanged record is ignored.
4.0	Verify Results	Transform History and the Import Log show total, inserts, updates, ignored, skipped and error counts for every load.

Ref.	Name	Description
5.0	Reports & Dashboard	Pie, bar and list reports are built on the target table and combined on the Employee Analytics Dashboard.
D1	Employee Import (u_employee_import)	Staging table holding raw spreadsheet rows before transformation.
D2	Employee Test (u_employee_test)	Target table holding final employee records.

Table 9: Processes and data stores

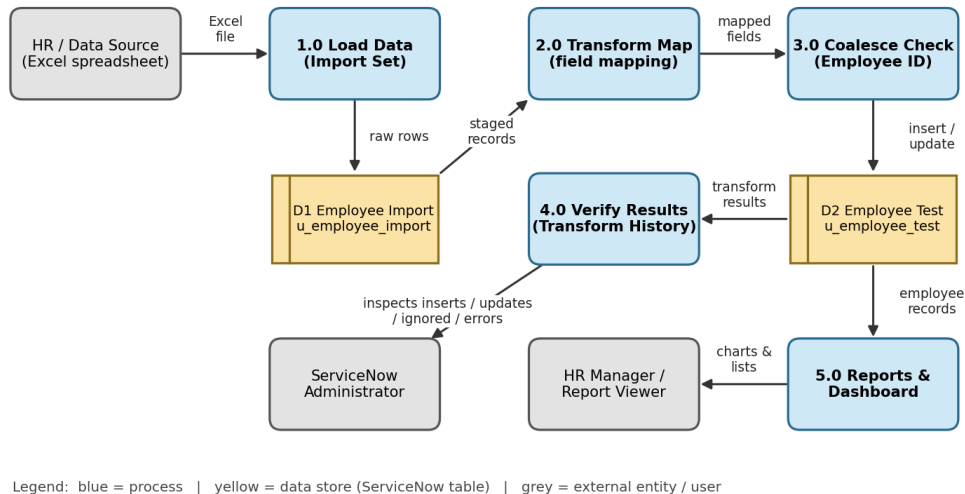
Data Flow Diagram (Level 1) - Import Data using Transform Maps

Figure 2: Data flow diagram (Level 1)

User stories

User type	Story	User story / task	Priority	Release
Administrator	USN-1	Create custom table Employee Test with the five fields	High	Sprint-1
Administrator	USN-2	Upload Sample Spreadsheet.xlsx using Load Data into Employee Import	High	Sprint-1
Administrator	USN-3	Create a Transform Map and auto-map the matching fields	High	Sprint-1
Administrator	USN-4	Run the Transform so staged records are created in the target table	High	Sprint-1
Administrator	USN-5	Enable Coalesce on Employee ID so repeat imports update instead of duplicating	High	Sprint-2
Administrator	USN-6	Open Transform History to see total, inserts, updates, ignored, skipped and errors	Medium	Sprint-2
HR Manager	USN-7	View a pie chart of employees by department	Medium	Sprint-2
HR Manager	USN-8	View a bar chart of employees by location	Medium	Sprint-2
HR Manager	USN-9	View a list report of all employees with five columns	Medium	Sprint-2
HR Manager	USN-10	See all three reports together on the Employee Analytics Dashboard	High	Sprint-2
Administrator	USN-11	Share the dashboard with selected users, groups and roles	Low	Sprint-2

Table 10: User stories

3.4 Technology Stack

Technical Architecture - Import Data using Transform Maps (Spreadsheet)

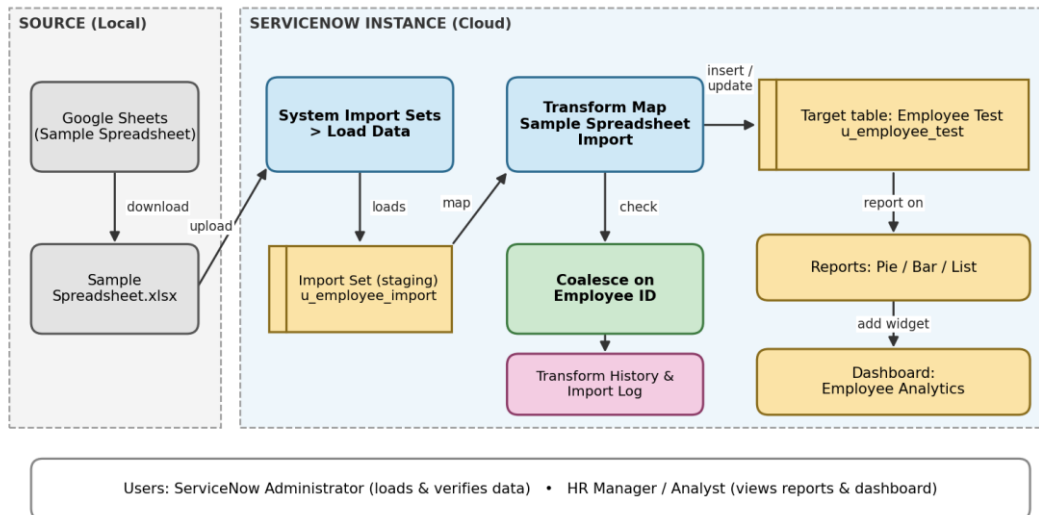


Figure 3: Technical architecture of the project

S.No	Component	Description	Technology
1	User Interface	ServiceNow web UI (navigator, forms, lists, Reports, Dashboards) and Google Sheets for source data	ServiceNow UI, Google Sheets
2	Application Logic-1 (Import)	Loads spreadsheet rows into a staging table before any real data changes	Load Data, table u_employee_import
3	Application Logic-2 (Transform)	Maps source fields to target fields and creates records	Transform Map "Sample Spreadsheet Import"
4	Application Logic-3 (Duplicate check)	Updates a record when the Employee ID already exists	Coalesce on Employee ID
5	Database	Custom target table with five String fields	ServiceNow table u_employee_test
6	Cloud Database / Infrastructure	All tables hosted by the ServiceNow cloud instance, used from any browser	ServiceNow cloud instance
7	File Storage	Spreadsheet used as the data source, uploaded once per load	Sample Spreadsheet.xlsx
8	External API / ML model	Not used: data is brought in by file upload	Not applicable

Table 11: Components and technologies

S.No	Characteristic	Description	Technology
1	Open-source frameworks	None needed; uses built-in low-code modules	Import Sets, Transform Maps, Reports, Dashboards
2	Security	Role-based access, ACLs and dashboard sharing restricted to users, groups and roles	Roles, ACLs, Share settings

S.No	Characteristic	Description	Technology
3	Scalable architecture	Staging table, Transform Map and target table are separate layers, so the map can be reused for larger files	3-tier ServiceNow architecture
4	Availability	Cloud hosted and browser accessed	ServiceNow cloud
5	Performance	Batch transform instead of manual entry: 15 rows in about 1.4 s and 4 rows in about 0.04 s	Import Set batch transform

Table 12: Application characteristics

4. PROJECT DESIGN

4.1 Problem Solution Fit

Problem-solution fit confirms that the problem found with the customer is actually solved by the solution. Here, the administrator's problems (slow manual entry, duplicate records after re-imports, and no clear view of employee data) are matched by a staged import, a reusable Transform Map, Coalesce on Employee ID and a reporting dashboard. The completed canvas is shown below.

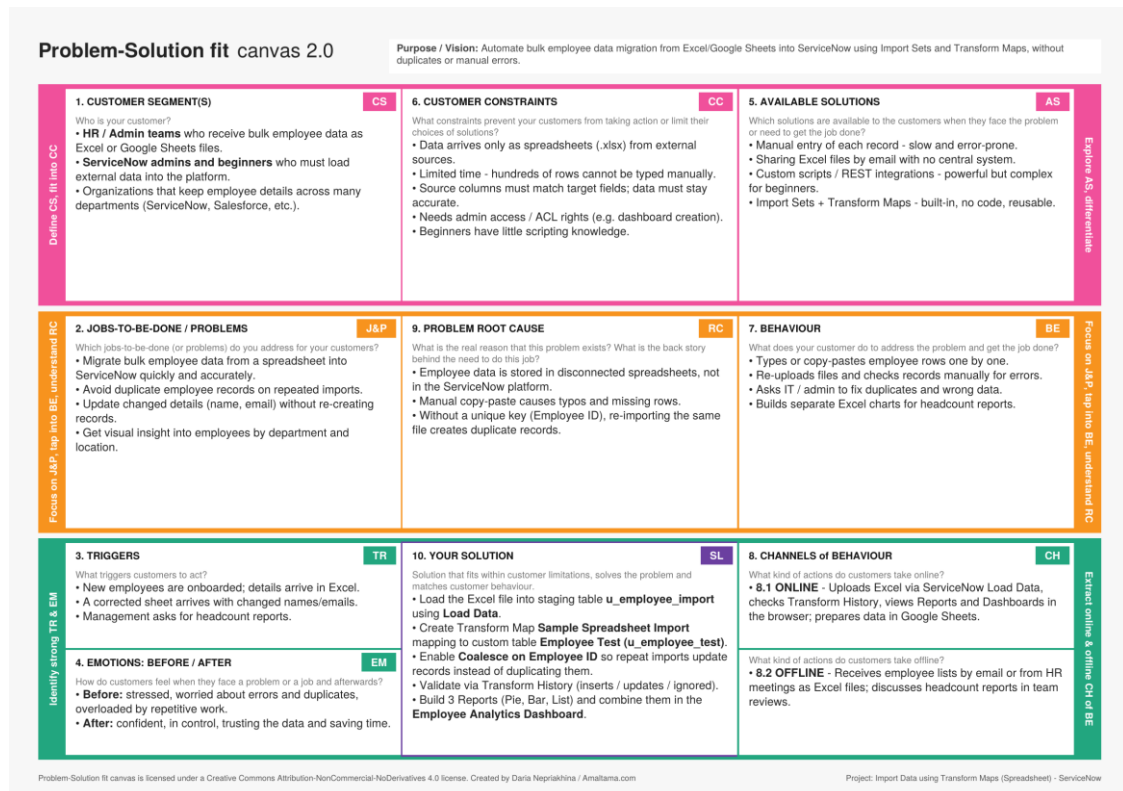


Figure 4: Completed problem-solution fit canvas

4.2 Proposed Solution

S.No	Parameter	Description
1	Problem statement	Bulk employee data arrives as Excel or Google Sheets files. Manual entry into ServiceNow is slow and error-prone, and importing the same file more than once creates duplicate records. There is no simple, reliable way for beginners and admins to migrate, update and monitor this data.
2	Idea / solution description	Use Import Sets and Transform Maps to load the spreadsheet into a staging table (u_employee_import) and map it to the custom table Employee Test (u_employee_test). Enable Coalesce on Employee ID so repeated imports update existing records and insert only new ones. Validate in Transform History, then build three reports and combine them in the Employee Analytics Dashboard.
3	Novelty / uniqueness	A completely no-code, end-to-end flow covering loading, mapping, transforming, de-duplicating, validating and reporting with built-in ServiceNow features only. The same Transform Map can be re-run with new files, and the project demonstrates insert, update and ignore behaviour on real sample data.

S.No	Parameter	Description
4	Social impact / customer satisfaction	Saves HR and admin teams hours of repetitive data entry, reduces human error and gives leadership trusted, up-to-date employee data. Beginners also gain a practical learning path for ServiceNow data import.
5	Business model (revenue model)	A reusable implementation accelerator: one-time setup fee, optional support and maintenance subscription, training workshops, and a packaged template (table, map, reports, dashboard) for other clients.
6	Scalability of the solution	Import Sets handle large row counts and any number of files. New fields, departments or target tables can be added by extending the Transform Map, and the approach extends to users, assets, locations or incidents. Scheduled imports and data sources (JDBC, REST, LDAP) can be added as volumes grow.

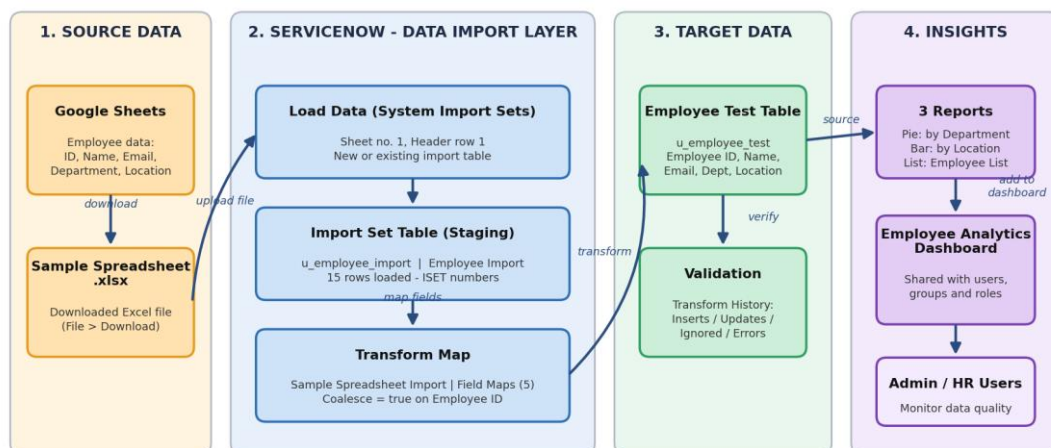
Table 13: Proposed solution

4.3 Solution Architecture

The architecture bridges the business problem and the technology solution. Data flows through five stages:

1. **Source data:** employee records are prepared in Google Sheets and downloaded as Sample Spreadsheet.xlsx.
2. **Load Data:** the file is uploaded through System Import Sets > Load Data (Sheet 1, Header row 1) into the staging table u_employee_import.
3. **Transform Map:** Sample Spreadsheet Import maps the five source fields to the target table Employee Test; Coalesce is enabled on Employee ID.
4. **Transform and validate:** running Transform inserts new records, updates matching ones and ignores unchanged rows; results are checked in Transform History.
5. **Reports and dashboard:** three reports (Pie, Bar, List) are added to the Employee Analytics Dashboard, which can be shared with users, groups and roles.

Solution Architecture - Import Data using Transform Maps (Spreadsheet) in ServiceNow



Coalesce on Employee ID: matching records are UPDATED, new IDs are INSERTED, identical rows are IGNORED - no duplicates.

Figure 5: Architecture and data flow of the project

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The work was planned in two Scrum sprints of six days each. An **Epic** is a large body of work, a **Story** is a small task within an epic, and a **Story Point** measures effort on a 1, 2, 3, 5 scale (1 = very easy, 2 = normal, 3 = moderate, 5 = difficult).

Product backlog and sprint plan

Sprint	Epic	Story	User story / task	Pts	Priority
Sprint-1	Data Preparation	USN-1	Create a Google Spreadsheet with sample employee data	1	High
Sprint-1		USN-2	Download the spreadsheet as Sample Spreadsheet.xlsx	1	High
Sprint-1	Custom Table	USN-3	Create the custom table Employee Test (u_employee_test)	2	High
Sprint-1		USN-4	Add the five String fields using Form Layout	2	High
Sprint-1	Import Set	USN-5	Load the Excel file into u_employee_import using Load Data	2	High
Sprint-1	Transform Map	USN-6	Create the Transform Map and map fields with Auto Map / Mapping Assist	3	High
Sprint-1		USN-7	Run the Transform and validate that 15 records are created	3	High
Sprint-2	Coalesce	USN-8	Enable Coalesce on Employee ID	2	High
Sprint-2		USN-9	Import updated data and verify 2 inserts and 2 updates	3	High
Sprint-2		USN-10	Re-import the same file and verify 4 rows are ignored	1	Medium
Sprint-2	Reports	USN-11	Report: Employees by Department (Pie, Count)	2	Medium
Sprint-2		USN-12	Report: Employees by Location (Bar, Count)	2	Medium
Sprint-2		USN-13	Report: Employee List (List, 5 columns)	2	Medium
Sprint-2	Dashboard	USN-14	Set Admin overrides on pa_dashboards ACL and create the dashboard	3	Medium
Sprint-2		USN-15	Add the three reports to the dashboard and share it	3	Medium

Table 14: Product backlog and sprint plan (total 32 story points)

Sprint schedule and velocity

Sprint	Story points	Duration	Start date	Planned end	Points completed	Release date
Sprint-1	14	6 days	19 Jan 2026	24 Jan 2026	14	24 Jan 2026
Sprint-2	18	6 days	26 Jan 2026	31 Jan 2026	18	31 Jan 2026

Table 15: Sprint schedule

Velocity: total story points completed = 14 + 18 = 32 across 2 sprints, so the team velocity is $32 / 2 = 16$ story points per sprint. With a six-day sprint, the average velocity is $16 / 6 \approx 2.67$ story points per day.

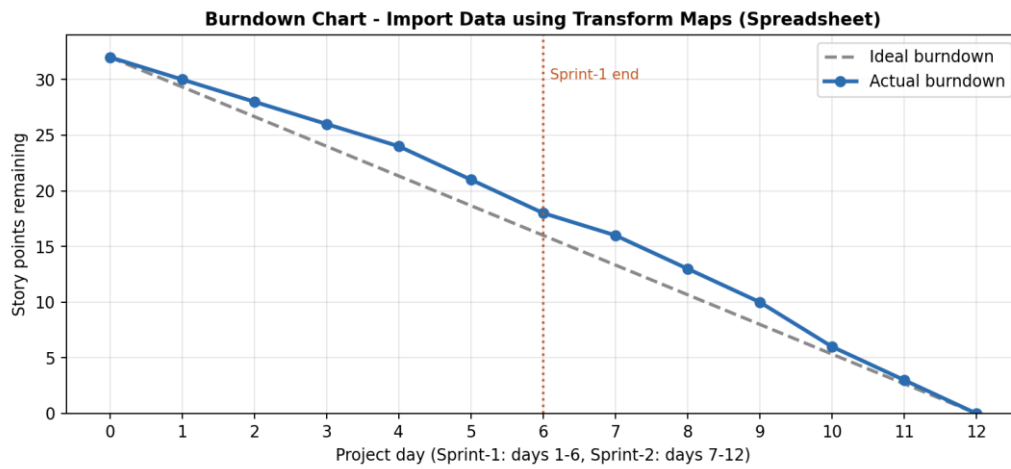


Figure 6: Burndown chart: ideal versus actual

The chart shows 32 story points remaining at the start, 18 at the end of Sprint-1 and 0 at the end of Sprint-2, closely following the ideal line.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Functional tests (FT), performance tests (PT) and user acceptance tests (UAT) were run between 19 and 31 January 2026 on the ServiceNow instance, covering user stories USN-1 to USN-15. All results below were observed during the project run.

Functional and performance test results

ID	Scenario	Expected result	Actual result	Result
FT-01	Data load (Load Data)	Table u_employee_import created, 15 rows loaded	State Complete, Success; Processed 15, inserts 15, errors 0	Pass
FT-02	Field mapping	All 5 source fields mapped	5 field maps created (u_name to u_employee_name, ID, department, location, email)	Pass
FT-03	Transform execution	Records created in Employee Test	Transformation complete, Success; 15 of 15 records visible	Pass
FT-04	Coalesce: update existing	Existing records updated, no duplicates	2 updates: SB-0004 name = Ajay; SB-0010 email = test18@gmail.com	Pass
FT-05	Coalesce: insert new	New IDs inserted	2 inserts: SB-0016 and SB-0017 (17 records)	Pass
FT-06	Duplicate prevention	No inserts or updates on re-import	Total 4, Inserts 0, Updates 0, Ignored 4, Errors 0	Pass
FT-07	Report output	Charts and list match table data	Pie: ServiceNow 11 / Salesforce 4 / AIML 2; Bar: Hyderabad 16 / Chennai 1; List: 17 rows	Pass
FT-08	Dashboard	All 3 reports display	List, Bar and Pie widgets shown; share options available	Pass
PT-01	Import response time	Small file loads in under 3 s	15 rows in about 1.4 s; 4 rows in 0.04 s	Pass
PT-02	Transform speed	Finishes within a few seconds	Run time 1 s for 4 rows; 0 s for the re-run	Pass
PT-03	Repeated import stability	No errors across repeated loads	3 imports (15, 4, 4 rows) completed with 0 errors	Pass

Table 16: Functional and performance test results

Import accuracy

Import	Rows	Inserts	Updates	Ignored	Errors
Import 1: initial file	15	15	0	0	0
Import 2: changed + new rows	4	2	2	0	0
Import 3: same 4 rows again	4	0	0	4	0
Total	23	17	2	4	0

Table 17: Import accuracy across three imports (100%, 0 errors)

User acceptance testing

Section	Total cases	Not tested	Fail	Pass
Data Preparation	1	0	0	1
Custom Table	2	0	0	2
Import Set (Load Data)	1	0	0	1
Transform Map	2	0	0	2
Coalesce (insert / update / ignore)	3	0	0	3
Reports	3	0	0	3
Dashboard	2	0	0	2
Total	14	0	0	14

Table 18: UAT test case analysis

Defects

Bug ID	Description	Severity	Status	Resolution / feedback
BG-001	Department typed as "Servicenow" (lower-case n) in updated rows SB-0004 and SB-0010, so it differs from "ServiceNow"	Low	Open	Correct the casing in the source sheet and re-import; Coalesce will update the records.
BG-002	Create Dashboard form not opening through PA_DASHBOARDS.FORM	Medium	Closed	Fixed by setting Admin overrides = true on the pa_dashboards create ACL.

Table 19: Bug tracking

7. RESULTS

7.1 Output Screenshots

The screenshots below show the main outputs of the project in the order the work was carried out.

The screenshot shows the 'Table Transform Map - New Record' form in ServiceNow. The form includes the following fields and values:

- Name: Sample Spreadsheet Import
- Source table: Employee Import [u_employee_import]
- Active: ☒
- Run business rules: ☒
- Enforce mandatory fields: No
- Copy empty fields: ☐
- Create new record on empty coalesce fields: ☐
- Application: Global
- Created: (empty)
- Target table: Employee Test [u_employee_test]
- Order: 100
- Run script: ☐

At the bottom left, there is a 'Submit' button and a 'Related Links' section with links for 'Auto Map Matching Fields' and 'Mapping Assist'.

Figure 7: Custom table Employee Test (u_employee_test) with its five String columns

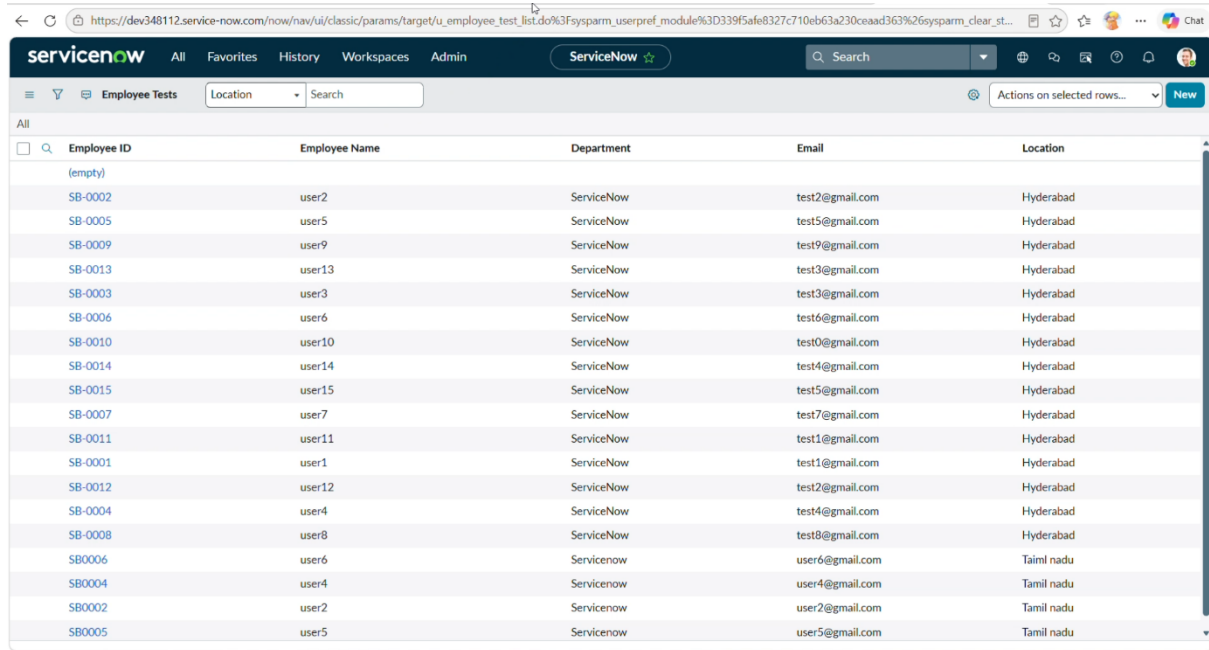
The screenshot shows the 'Import Processor' progress page in ServiceNow. The page displays the following information:

- Name: ImportProcessor
- State: Complete
- Completion code: Success
- Message: Processed: 15, inserts 15, updates 0, errors 0, empty and ignored 0, ignored errors 0 (0:00:00.532)

Below the progress section, there is a 'Next steps...' section with the following links:

- Import sets: Go to the import sets for this data load
- Loaded data: Go to the newly imported data inside the staging table: u_employee_import
- Create transform map: Create a transform map for the newly staged data
- Run Transform: Transform a loaded import set using an existing transform map
- Import log: View the import log

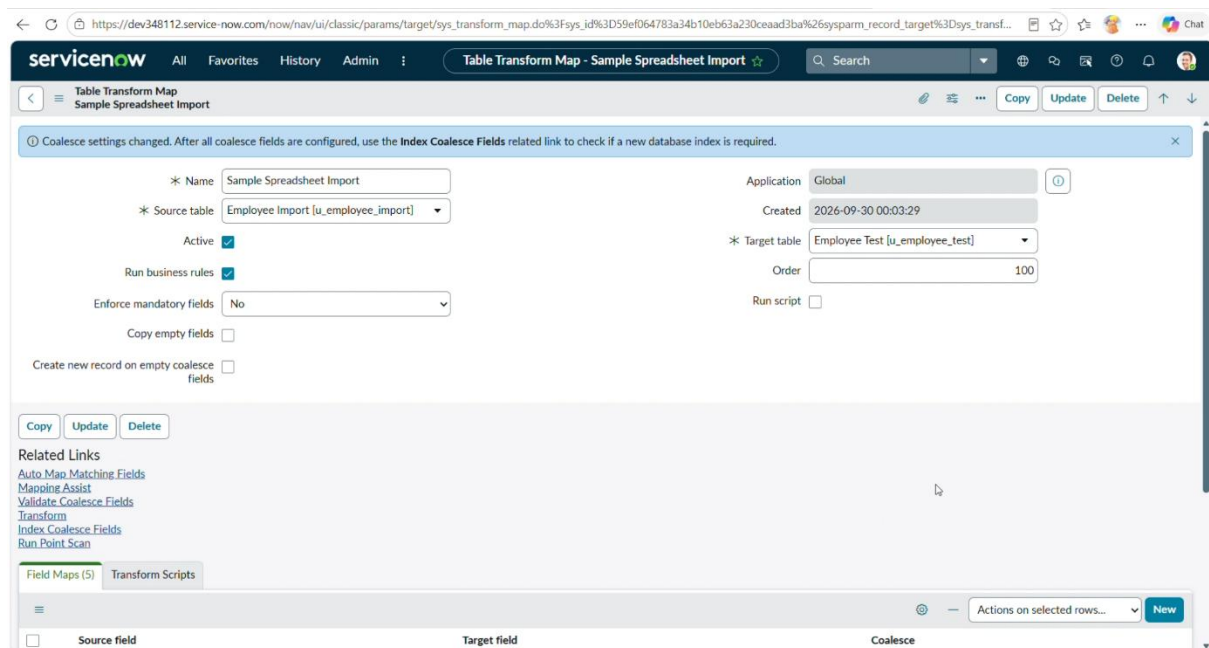
Figure 8: Load Data result: State Complete, Success; 15 processed, 15 inserts, 0 errors



The screenshot shows the ServiceNow interface for the 'Employee Tests' list. The table displays 15 records with columns: Employee ID, Employee Name, Department, Email, and Location. The records are sorted by Employee ID in descending order.

Employee ID	Employee Name	Department	Email	Location
SB-0002	user2	ServiceNow	test2@gmail.com	Hyderabad
SB-0005	user5	ServiceNow	test5@gmail.com	Hyderabad
SB-0009	user9	ServiceNow	test9@gmail.com	Hyderabad
SB-0013	user13	ServiceNow	test3@gmail.com	Hyderabad
SB-0003	user3	ServiceNow	test3@gmail.com	Hyderabad
SB-0006	user6	ServiceNow	test6@gmail.com	Hyderabad
SB-0010	user10	ServiceNow	test0@gmail.com	Hyderabad
SB-0014	user14	ServiceNow	test4@gmail.com	Hyderabad
SB-0015	user15	ServiceNow	test5@gmail.com	Hyderabad
SB-0007	user7	ServiceNow	test7@gmail.com	Hyderabad
SB-0011	user11	ServiceNow	test1@gmail.com	Hyderabad
SB-0001	user1	ServiceNow	test1@gmail.com	Hyderabad
SB-0012	user12	ServiceNow	test2@gmail.com	Hyderabad
SB-0004	user4	ServiceNow	test4@gmail.com	Hyderabad
SB-0008	user8	ServiceNow	test8@gmail.com	Hyderabad
SB0006	user6	ServiceNow	user6@gmail.com	Tamil nadu
SB0004	user4	ServiceNow	user4@gmail.com	Tamil nadu
SB0002	user2	ServiceNow	user2@gmail.com	Tamil nadu
SB0005	user5	ServiceNow	user5@gmail.com	Tamil nadu

Figure 9: Employee Tests list showing all 15 imported records



The screenshot shows the configuration page for a Transform Map named 'Sample Spreadsheet Import'. The configuration includes settings for the source table, target table, and various options.

Configuration Details:

- Name:** Sample Spreadsheet Import
- Source table:** Employee Import [u_employee_import]
- Active:** ☒
- Run business rules:** ☒
- Enforce mandatory fields:** No
- Copy empty fields:** ☐
- Create new record on empty coalesce fields:** ☐
- Application:** Global
- Created:** 2026-09-30 00:03:29
- Target table:** Employee Test [u_employee_test]
- Order:** 100
- Run script:** ☐

Related Links:

- [Auto Map Matching Fields](#)
- [Mapping Assist](#)
- [Validate Coalesce Fields](#)
- [Transform](#)
- [Index Coalesce Fields](#)
- [Run Point Scan](#)

Field Maps (5):

Source field	Target field	Coalesce

Figure 10: Transform Map "Sample Spreadsheet Import" from Employee Import to Employee Test

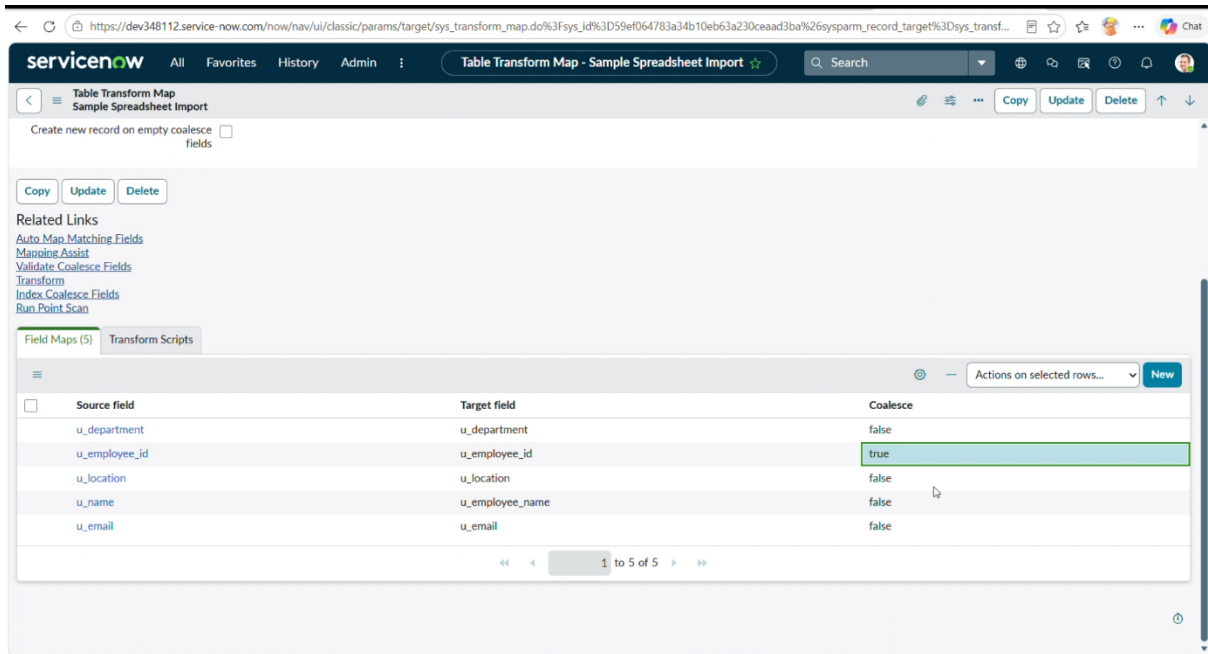


Figure 11: Field maps with Coalesce set to true on Employee ID

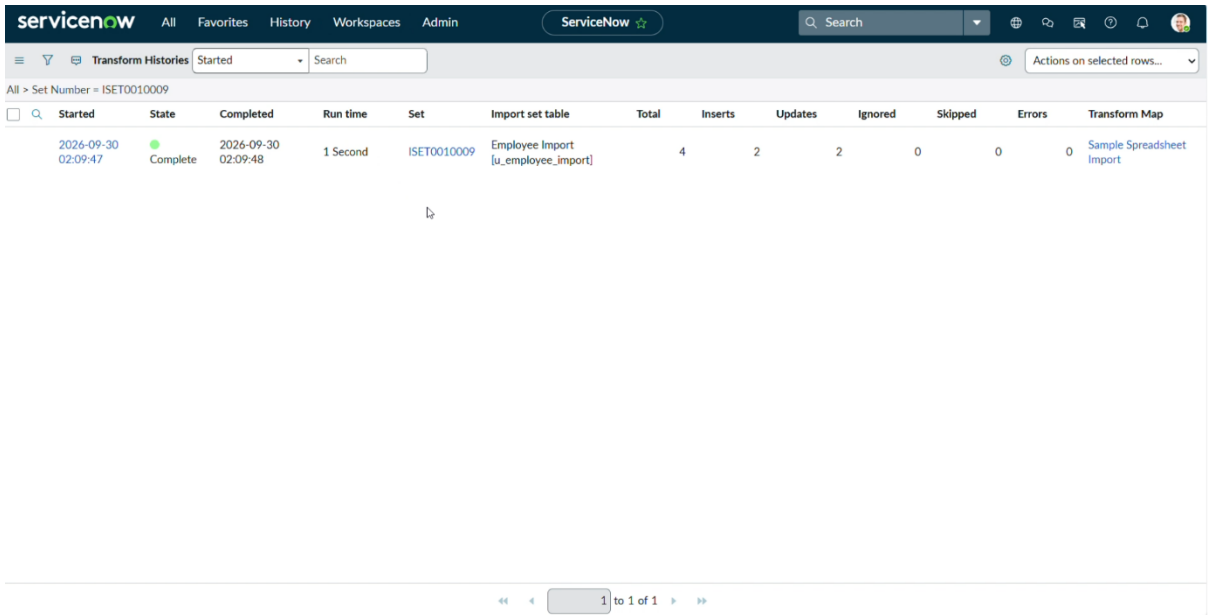


Figure 12: Transform History for the updated file: Total 4, Inserts 2, Updates 2

Employee ID	Employee Name	Department	Email	Location
(empty)				
SB-0002	user2	ServiceNow	test2@gmail.com	Hyderabad
SB-0005	user5	ServiceNow	test5@gmail.com	Hyderabad
SB-0009	user9	ServiceNow	test9@gmail.com	Hyderabad
SB-0013	user13	ServiceNow	test3@gmail.com	Hyderabad
SB-0003	user3	ServiceNow	test3@gmail.com	Hyderabad
SB-0006	user6	ServiceNow	test6@gmail.com	Hyderabad
SB-0010	user10	ServiceNow	test0@gmail.com	Hyderabad
SB-0014	user14	ServiceNow	test4@gmail.com	Hyderabad
SB-0015	user15	ServiceNow	test5@gmail.com	Hyderabad
SB-0007	user7	ServiceNow	test7@gmail.com	Hyderabad
SB-0011	user11	ServiceNow	test1@gmail.com	Hyderabad
SB-0001	user1	ServiceNow	test1@gmail.com	Hyderabad
SB-0012	user12	ServiceNow	test2@gmail.com	Hyderabad
SB-0004	user4	ServiceNow	test4@gmail.com	Hyderabad
SB-0008	user8	ServiceNow	test8@gmail.com	Hyderabad
SB0006	user6	ServiceNow	user6@gmail.com	Tamil nadu
SB0004	user4	ServiceNow	user4@gmail.com	Tamil nadu
SB0002	user2	ServiceNow	user2@gmail.com	Tamil nadu
SB0005	user5	ServiceNow	user5@gmail.com	Tamil nadu

Figure 14: Employee Tests list with 17 records after the update and new inserts (SB-0004, SB-0010, SB-0016, SB-0017)

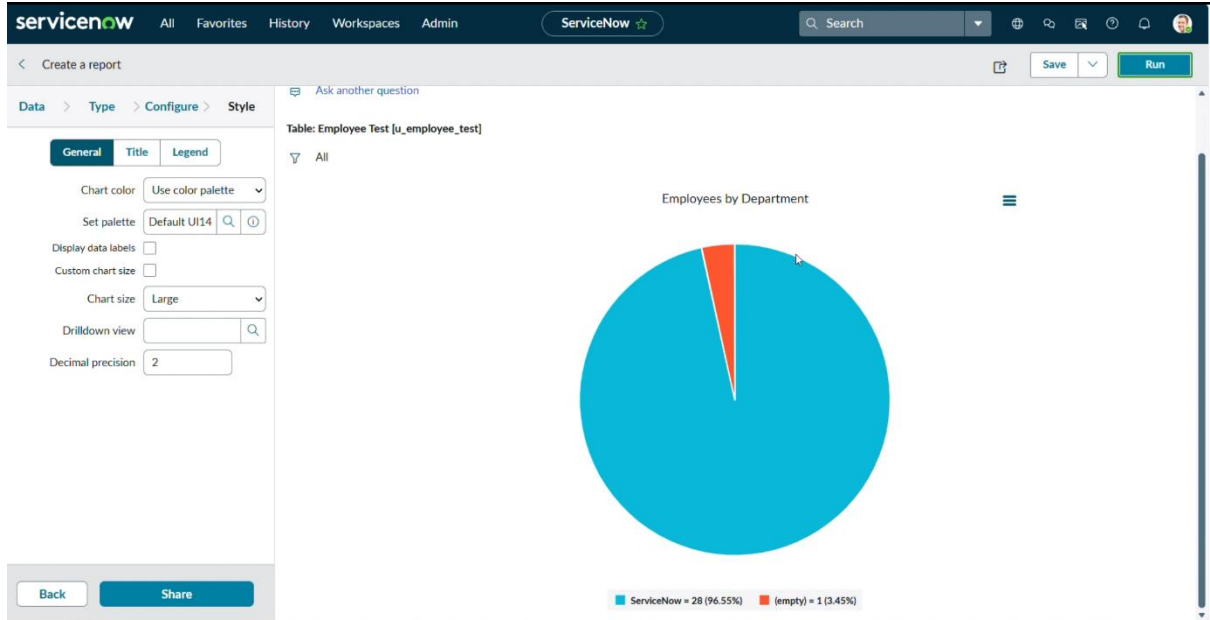


Figure 15: Report 1: Employees by Department (Pie): ServiceNow 11, Salesforce 4, AIML 2

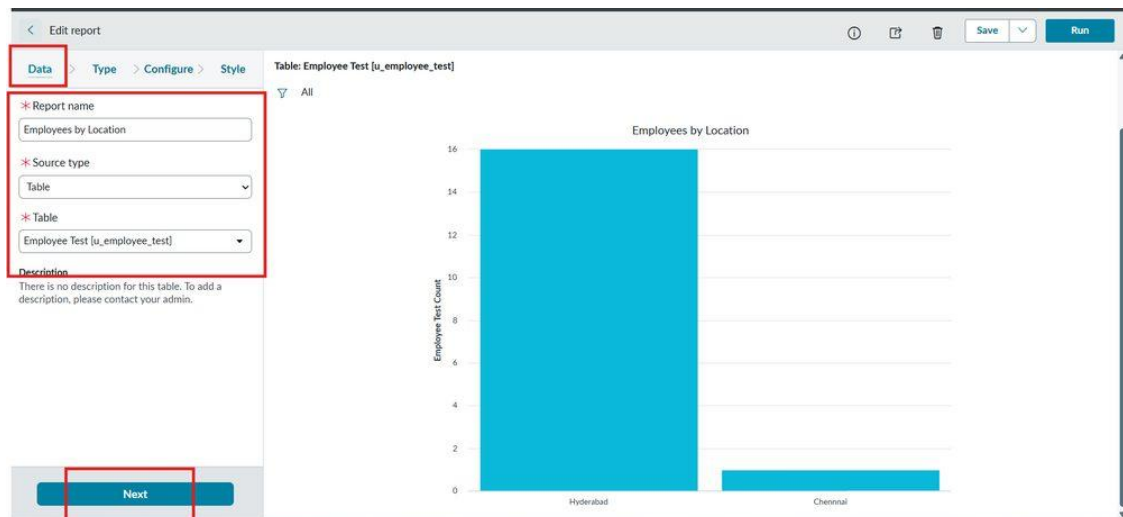


Figure 16: Report 2: Employees by Location (Bar): Hyderabad 16, Chennai 1

The screenshot shows the 'Create report' window for a report titled 'Employees list table'. The 'Data' tab is active, showing a table with 5 columns: Employee ID, Employee Name, Email, Department, and Location. The table contains 17 rows of data. The interface also includes a 'Choose columns' section and a 'Group by' dropdown menu.

Employee ID	Employee Name	Email	Department	Location
SB-0009	user9	test9@gmail.com	ServiceNow	Hyderabad
SB-0013	user13	test3@gmail.com	ServiceNow	Hyderabad
SB-0003	user3	test3@gmail.com	ServiceNow	Hyderabad
SB-0006	user6	test6@gmail.com	ServiceNow	Hyderabad
SB-0010	user10	test0@gmail.com	ServiceNow	Hyderabad
SB-0014	user14	test4@gmail.com	ServiceNow	Hyderabad
SB-0015	user15	test5@gmail.com	ServiceNow	Hyderabad
SB-0007	user7	test7@gmail.com	ServiceNow	Hyderabad

Figure 17: Report 3: Employee List Report (List with five columns, 17 rows)

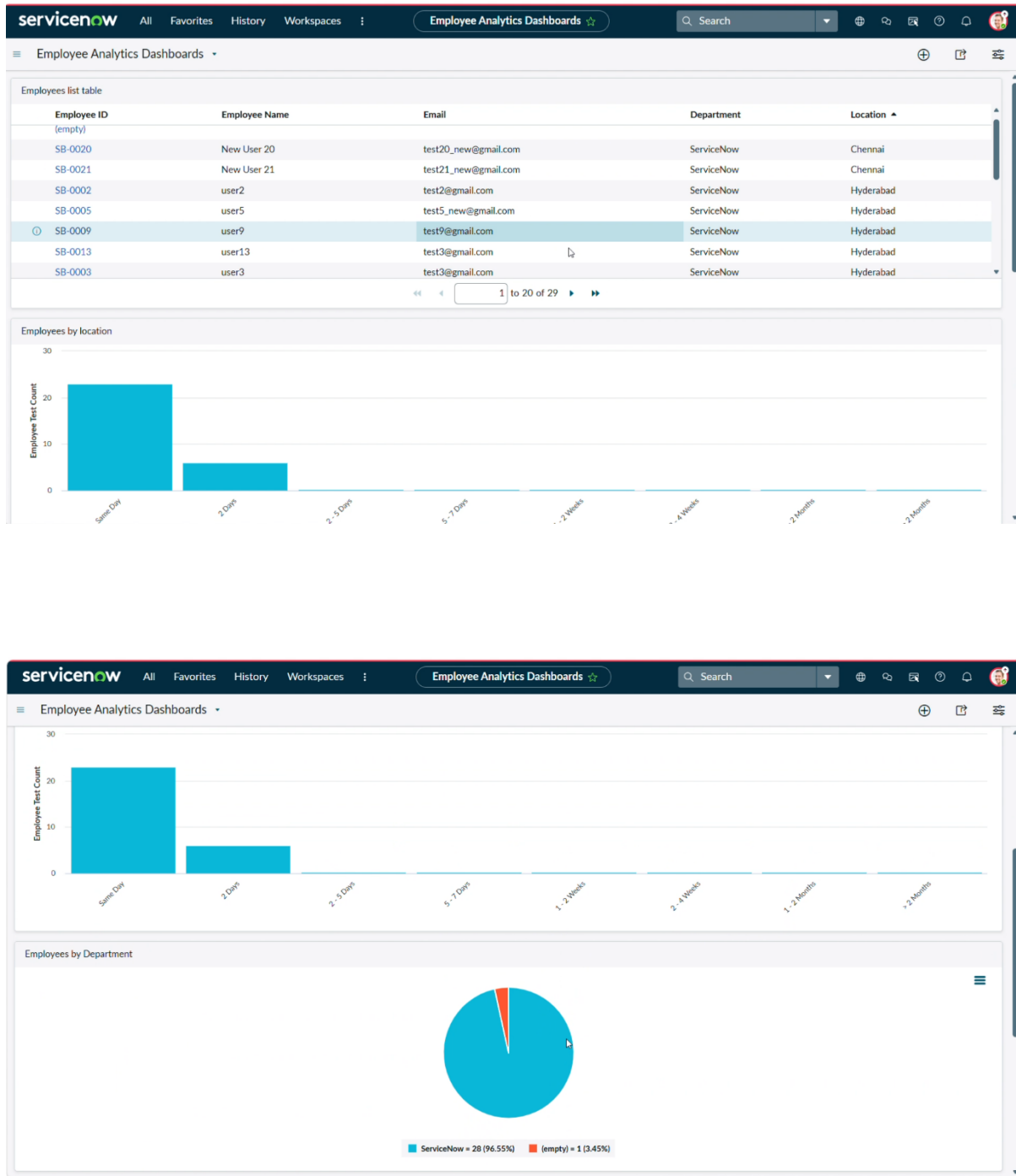


Figure 18: Employee Analytics Dashboard with all three reports and the Share panel

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Saves time:** a whole spreadsheet is loaded and transformed in about a second instead of typing records one by one.
- **No duplicates:** Coalesce on Employee ID updates existing employees and inserts only new ones; re-importing the same file changed nothing.
- **Safe staging:** raw data is held in an Import Set table, so live data is touched only after mapping.
- **Reusable:** the same Transform Map can be run again for every new spreadsheet.
- **Traceable:** Transform History and the Import Log show inserts, updates, ignored rows and errors for every load.
- **Visible results:** reports and a shareable dashboard give HR a clear view by department and location.
- **No code and no extra cost:** everything uses built-in ServiceNow features and a browser.

Disadvantages

- **Manual trigger:** each file must be uploaded and transformed by an administrator; imports are not yet scheduled.
- **Data quality depends on the source:** typing differences such as “Servicenow” versus “ServiceNow” (BG-001) are imported as they are.
- **Single match key:** only Employee ID is used for Coalesce, so a wrongly typed ID would create a new record.
- **Small sample:** the project was tested with 15 to 17 rows, so very large volumes were not measured.
- **Admin access needed:** creating tables, Transform Maps and the dashboard form requires administrator rights and an ACL change.
- **Basic validation:** mandatory-field rules and data-cleaning scripts were not implemented.

9. CONCLUSION

This project successfully implemented an automated employee data management solution using ServiceNow Import Sets and Transform Maps, ensuring accuracy, efficiency and data integrity throughout the employee onboarding and update process. Fifteen employee records were loaded and transformed into the custom Employee Test table without errors, and later imports added two new employees and updated two existing ones.

A key enhancement was the use of Coalesce on the Employee ID field. It lets the system identify existing employee records on every import, which prevents duplicates and makes repeated imports update existing data instead of inserting redundant entries. Re-importing the same file produced 0 inserts, 0 updates and 4 ignored rows, which confirmed that the employee table stays clean, reliable and consistent.

The project was extended with three reports (Employees by Department, Employees by Location and Employee List) and the Employee Analytics Dashboard, giving administrators and HR teams a single view of employee data and its distribution. All 14 user acceptance test cases and all functional and performance tests passed, and the two sprints delivered 32 story points at a velocity of 16 points per sprint. Only one low-severity data-casing observation (BG-001) remains open. Together, these results show that the approach is practical, repeatable and suitable for real enterprise data migration.

10. FUTURE SCOPE

Several ideas from the brainstorming session were kept outside the current scope and can be added in the future:

- **More file formats:** accept CSV and TSV files in addition to .xlsx.
- **Scheduled imports:** connect an external Data Source (JDBC, REST or LDAP) and run the import automatically on a schedule.
- **Data cleaning:** add transform scripts to trim spaces, fix letter case (for example “Servicenow”) and standardise names before records are inserted.
- **Validation rules:** enforce mandatory fields and value checks before insertion.
- **Import reporting:** add reports for import success and error counts, newly added employees and updated employees.
- **Scheduled and shared reports:** email reports to HR and managers automatically.
- **Wider use:** reuse the same pattern to import users, assets, locations or incidents.
- **Notifications:** alert administrators when an import finishes with errors.

11. APPENDIX

Dataset

The dataset is a Google Sheet exported as **Sample Spreadsheet.xlsx** (Sheet 1, Header row 1) with five columns. The initial file contains 15 employees.

Employee ID	Name	Email	Department	Location
SB-0001	user1	test1@gmail.com	ServiceNow	Hyderabad
SB-0002	user2	test2@gmail.com	ServiceNow	Hyderabad
SB-0003	user3	test3@gmail.com	ServiceNow	Hyderabad
SB-0004	user4	test4@gmail.com	ServiceNow	Hyderabad
SB-0005	user5	test5@gmail.com	ServiceNow	Hyderabad
SB-0006	user6	test6@gmail.com	ServiceNow	Hyderabad
SB-0007	user7	test7@gmail.com	ServiceNow	Hyderabad
SB-0008	user8	test8@gmail.com	ServiceNow	Hyderabad
SB-0009	user9	test9@gmail.com	ServiceNow	Hyderabad
SB-0010	user10	test0@gmail.com	ServiceNow	Hyderabad
SB-0011	user11	test1@gmail.com	ServiceNow	Hyderabad
SB-0012	user12	test2@gmail.com	ServiceNow	Hyderabad
SB-0013	user13	test3@gmail.com	ServiceNow	Hyderabad
SB-0014	user14	test4@gmail.com	ServiceNow	Hyderabad
SB-0015	user15	test5@gmail.com	ServiceNow	Hyderabad

Table 20: Initial dataset (Import 1, 15 rows)

Employee ID	Name	Email	Department	Location
SB-0004	Ajay	test4_new@gmail.com	ServiceNow	Hyderabad
SB-0005	user5	test5_new@gmail.com	ServiceNow	Hyderabad
SB-0020	New User 20	test20_new@gmail.com	ServiceNow	Chennai
SB-0021	New User 21	test21_new@gmail.com	ServiceNow	Chennai

Table 21: Updated dataset (Import 2, 4 rows; re-imported unchanged as Import 3)

Configuration Summary

Object	Configuration
Target table	Label: Employee Test; Name: u_employee_test; fields (String): Employee ID, Employee Name, Email, Department, Location
Import Set table	Label: Employee Import; Name: u_employee_import (auto-populated); created through Load Data

Object	Configuration
Transform Map	Name: Sample Spreadsheet Import; Source: Employee Import; Target: Employee Test; Order 100; Run business rules: true; Enforce mandatory fields: No
Field maps (5)	u_name to u_employee_name; u_employee_id to u_employee_id (Coalesce = true); u_department to u_department; u_location to u_location; u_email to u_email
ACL	pa_dashboards, operation create: Admin overrides = true (needed to open PA_DASHBOARDS.FORM)
Reports	Employees by Department (Pie, Group by Department, Count); Employees by Location (Bar, Group by Location, Count); Employee List Report (List, 5 columns)
Dashboard	Employee Analytics Dashboard with the three reports; sharing by users, groups and roles

Table 22: Configuration summary

Source Code

No custom source code was written. The solution is configured entirely with ServiceNow out-of-the-box features (Tables, Import Sets, Transform Maps, Reports, Dashboards and ACLs).

GitHub & Project Demo Link

GitHub repository	monigem68-sys/NM-project
Project demo video	https://drive.google.com/file/d/1XFWwOf481fCWBD4qis68xswjLs1rnlcR/view?usp=drivesdk
ServiceNow instance	https://dev348112.service-now.com

References

- ServiceNow Product Documentation: Import sets, Transform maps and Coalesce (docs.servicenow.com)
- ServiceNow Product Documentation: Performance Analytics dashboards and Reports
- Atlassian Agile Coach: Scrum sprints, story points and burndown charts (atlassian.com/agile)
- Miro / Mural templates: Customer problem statement, empathy map, brainstorming and idea prioritisation